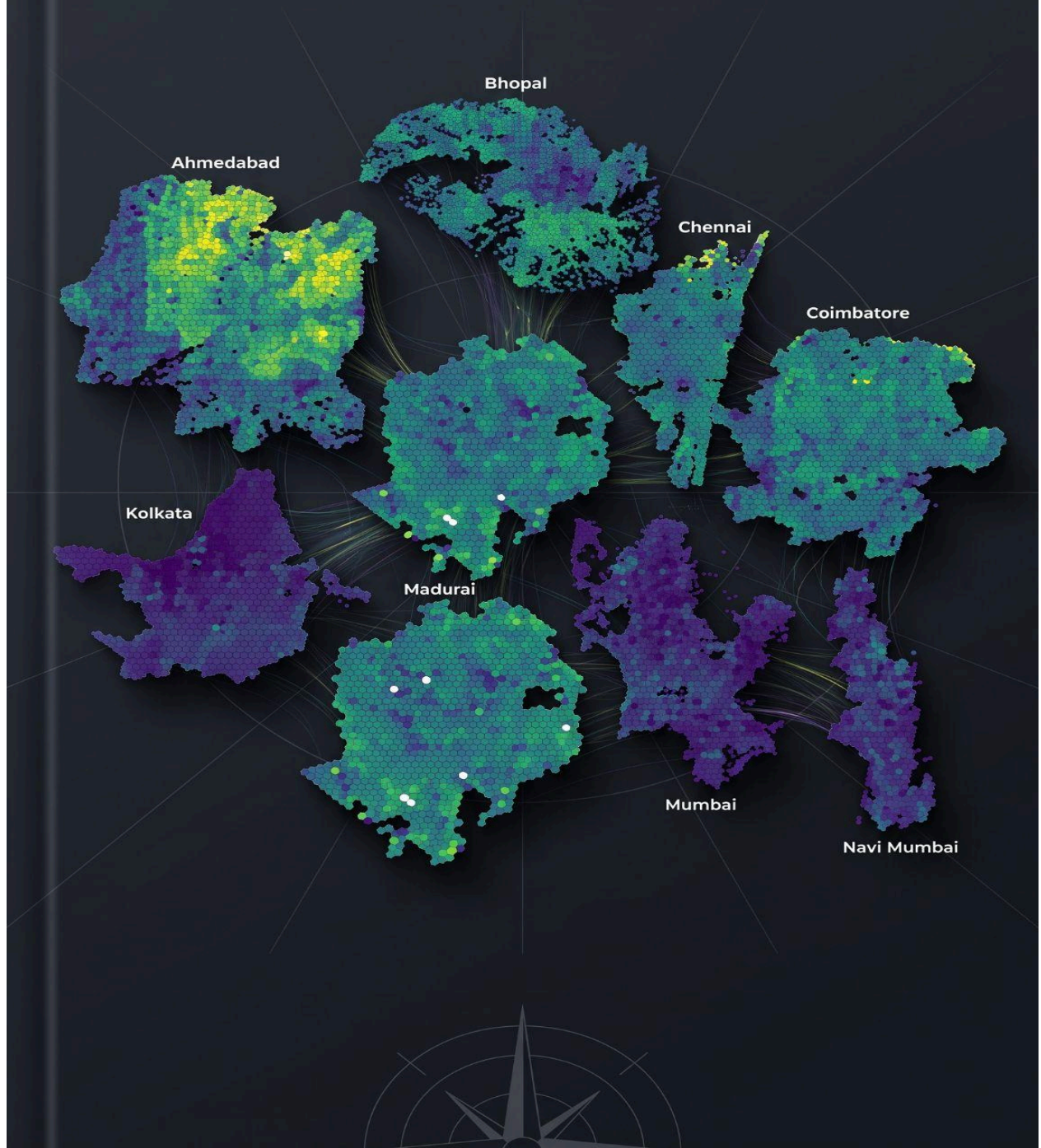


From Rooftops to Resilience: A Geospatial Framework for Mapping Rooftop Albedo Across Indian Cities



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Abstract:

Urban rooftops play an important role in climate adaptation, urban heat mitigation, and the transition toward sustainable cities. This study presents a reproducible geospatial framework for analyzing rooftop albedo across eight major Indian cities using Google's Rooftop Albedo dataset. More than 3.34 million rooftop observations were processed through data quality assessment, OpenStreetMap boundary integration, H3 hexagonal spatial aggregation, and statistical analysis to generate standardized maps and comparative metrics of rooftop reflectivity.

The analysis revealed significant spatial variability both within and among cities. Ahmedabad exhibited the highest mean rooftop albedo (0.239), while Kolkata recorded the lowest (0.132), highlighting considerable differences in rooftop surface characteristics across urban environments. Rather than estimating rooftop solar potential, this work establishes an open-source methodological framework that can support future studies on cool roofs, urban heat island mitigation, distributed renewable energy, and climate-resilient urban planning.

Keywords: Rooftop Albedo, Geospatial Analysis, Urban Climate, Google Rooftop Albedo Dataset, Sustainable Cities.

Introduction:

India's rapid urbanization and increasing energy demand have intensified the need for sustainable urban development strategies. While large-scale renewable energy infrastructure has significantly accelerated India's clean energy transition, it has also raised important questions regarding land use, ecological conservation, and community livelihoods.

Urban rooftops represent one of the largest underutilized assets within cities. Beyond their potential for decentralized renewable energy generation, rooftops influence urban temperatures through their surface reflectivity (albedo), making them an important component of climate adaptation and heat mitigation strategies.

This project develops a reproducible geospatial framework to analyze rooftop albedo across major Indian cities using Google's Rooftop Albedo Dataset. Rather than focusing on individual buildings, the study establishes a scalable methodology for characterizing rooftop reflectivity at the city scale through open-source geospatial tools and hierarchical spatial indexing.

Research Motivation:

The motivation behind this work extends beyond mapping rooftop characteristics.

Across India, the expansion of utility-scale renewable energy infrastructure has transformed many rural landscapes. While these projects contribute significantly to national renewable energy targets, they also raise broader questions concerning agricultural land conversion, ecological impacts, and local livelihoods.

This project explores an alternative perspective:

How can existing urban infrastructure contribute to India's clean energy transition while simultaneously supporting urban climate resilience?

Although this study does not estimate rooftop solar capacity or compare rooftop systems directly with utility-scale solar parks, it establishes an essential first step by creating a standardized spatial framework for understanding rooftop characteristics across Indian cities.

Objectives:

The primary objectives of this study were to:

1. Develop a reproducible workflow for processing Google's Rooftop Albedo Dataset.
2. Perform quality assessment and preprocessing of rooftop observations.
3. Delineate administrative boundaries for selected Indian cities using OpenStreetMap.
4. Aggregate rooftop observations using the H3 hierarchical spatial indexing system.
5. Compare rooftop albedo characteristics across multiple Indian cities.
6. Produce publication-quality spatial visualizations and statistical summaries.

Study Area:

Eight major Indian cities representing diverse climatic conditions and urban morphologies were selected:

1. Ahmedabad

2. Bhopal
3. Chennai
4. Coimbatore
5. Kolkata
6. Madurai
7. Mumbai
8. Navi Mumbai

Data Sources:

Google Rooftop Albedo Dataset

The primary dataset consists of rooftop-level albedo observations derived from Google's open rooftop mapping initiative.

Variables include:

- Mean rooftop albedo
- Latitude
- Longitude

The dataset contained approximately 3.34 million rooftop observations across the selected cities.

Administrative Boundaries:

Administrative city boundaries were retrieved from OpenStreetMap using the OSMnx Python library to ensure consistent spatial delineation.

Methodology:

The analytical workflow consists of six major stages:

1. Data Quality Assessment

The rooftop dataset was screened to remove:

- Missing observations
- Duplicate records
- Invalid geographic coordinates
- Physically unrealistic albedo values

2. Geospatial Processing

The cleaned dataset was converted into a GeoDataFrame using the WGS84 coordinate reference system (EPSG:4326).

3. Administrative Boundary Integration:

City boundaries were downloaded from OpenStreetMap and validated before analysis.

4. Spatial Clipping:

Rooftop observations were clipped to each administrative boundary, producing independent datasets for every study city.

5. H3 Spatial Aggregation:

Millions of rooftop observations were aggregated into standardized H3 hexagonal cells.

This approach:

- reduces visual clutter,
- preserves neighborhood-scale variability,
- enables consistent comparison between cities,
- produces scalable spatial datasets.

6. Statistical Analysis:

For each city, descriptive statistics including:

- Mean
- Median
- Minimum
- Maximum
- Standard Deviation
- Quartiles

were computed to characterize rooftop reflectivity.

Key Results:

The proposed geospatial workflow successfully processed 3.34 million rooftop observations from Google's Rooftop Albedo dataset and transformed them into standardized H3 hexagonal

spatial products across eight major Indian cities. After quality assessment, only 28 observations ($<0.001\%$) were removed, demonstrating the high completeness and reliability of the dataset for urban-scale geospatial analysis.

Comparative analysis revealed substantial differences in rooftop reflectivity among cities. Ahmedabad exhibited the highest mean rooftop albedo (0.239), followed by Coimbatore (0.217) and Bhopal (0.210), whereas Kolkata (0.132), Mumbai (0.139), and Navi Mumbai (0.144) recorded the lowest average rooftop albedo values. These results indicate pronounced inter-city variability in rooftop surface characteristics.

The H3 spatial aggregation approach effectively transformed millions of individual rooftop observations into neighborhood-scale representations, revealing considerable spatial heterogeneity within every city. Even cities with similar average albedo values displayed distinct local patterns, emphasizing that rooftop reflectivity cannot be adequately characterized using city-wide averages alone.

Distributional analysis further demonstrated clear differences in rooftop albedo composition. Ahmedabad contained the highest proportion of moderate-to-high albedo rooftops, with approximately 76% of rooftops exhibiting albedo values greater than 0.20. In contrast, more than 95% of rooftops in Mumbai and nearly 98.5% in Kolkata fell within the low-albedo (<0.20) category, indicating predominantly darker rooftop surfaces.

Overall, the study establishes a fully reproducible, open-source geospatial workflow for processing, aggregating, and comparing rooftop albedo across Indian cities. The framework provides a robust methodological foundation for future investigations into urban heat mitigation, cool-roof strategies, rooftop photovoltaic assessment, and climate-resilient urban planning.

Significance:

This work should be viewed as a methodological framework rather than a direct assessment of rooftop solar potential.

The framework establishes a reproducible foundation that can be extended by integrating additional datasets such as:

1. Rooftop geometry

2. Solar irradiance
3. Land Surface Temperature
4. Building footprints
5. Population density
6. Electricity demand
7. Urban land cover

Such integrations could support future studies on:

1. Cool-roof implementation
2. Urban heat island mitigation
3. Rooftop photovoltaic potential
4. Climate-resilient city planning
5. Distributed renewable energy systems

Limitations:

1. Only eight cities were analyzed.
2. Building footprints were not incorporated.
3. Rooftop orientation and geometry were not considered.
4. Rooftop albedo is not equivalent to solar suitability.
5. Temporal variability was not assessed.

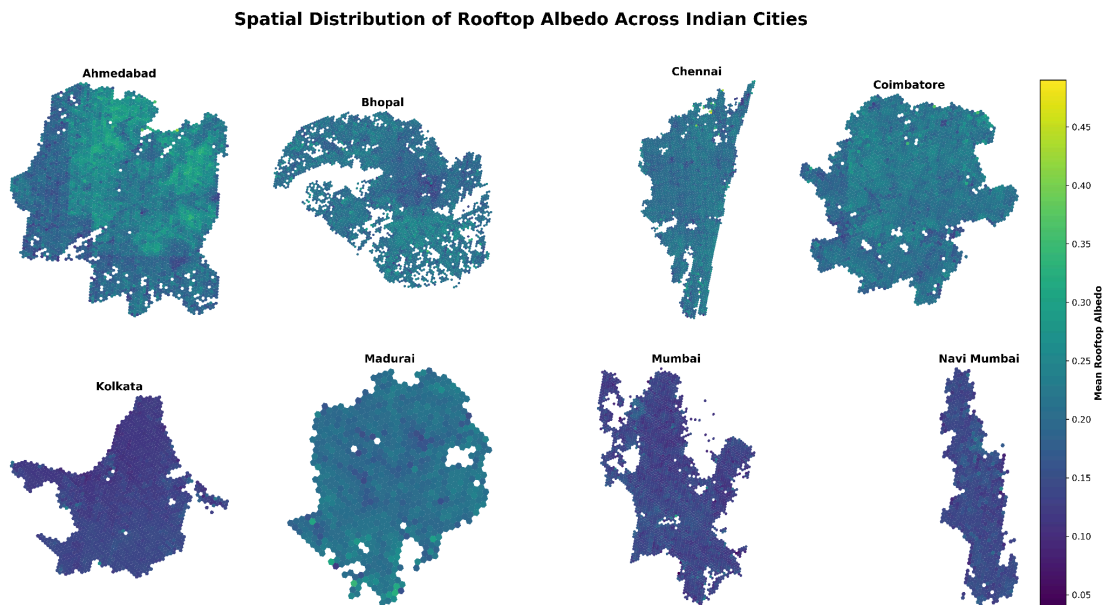
Acknowledgements:

This project builds upon the availability of open geospatial datasets and open-source scientific software that make reproducible urban climate research possible.

I would like to acknowledge *Milan Janosov* for openly sharing his rooftop albedo workflow and methodology through his public article and accompanying notebook. His work provided valuable methodological guidance for processing and visualizing Google's Rooftop Albedo dataset. The present study adapts and substantially extends those concepts to develop a reproducible framework for comparative rooftop albedo analysis across multiple Indian cities using H3 spatial indexing and open-source geospatial tools.

Figures and Interpretations:

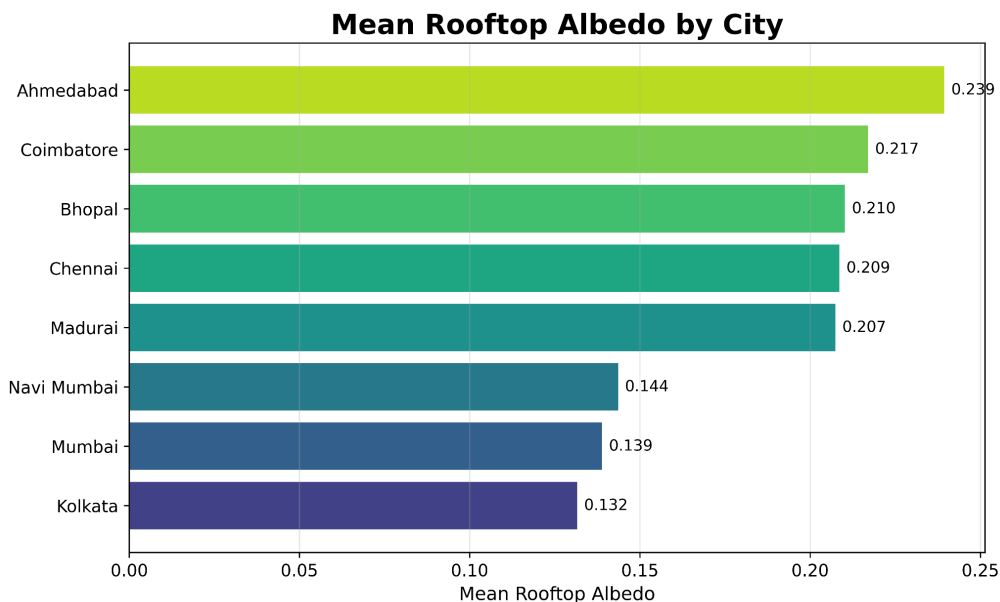
Figure.1:
Spatial Distribution of Rooftop Albedo Across Indian Cities:



Interpretation:

The H3 aggregated maps reveal considerable spatial heterogeneity in rooftop reflectivity across the eight study cities. Ahmedabad exhibits the highest concentration of relatively reflective rooftops, whereas Mumbai, Navi Mumbai, and Kolkata display predominantly lower rooftop albedo values. The neighborhood-scale variability observed within each city highlights the importance of localized analysis rather than relying solely on city-wide averages.

Figure 2:
Mean Rooftop Albedo Ranking:



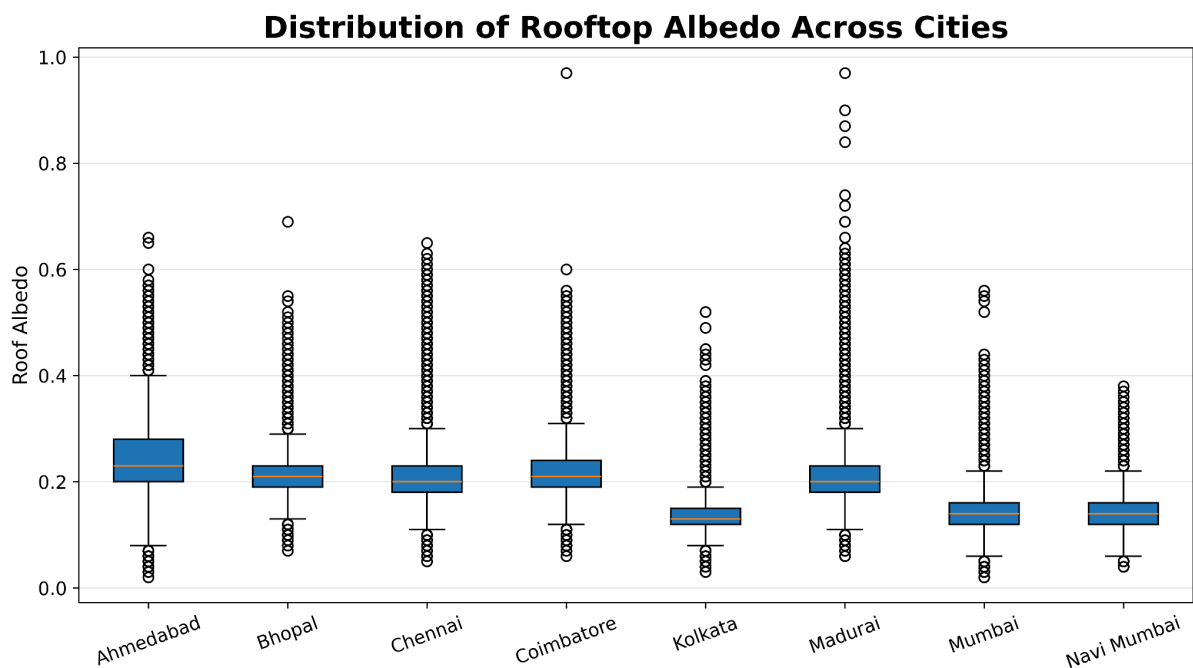
Interpretation:

The comparative ranking demonstrates that Ahmedabad recorded the highest mean rooftop albedo (0.239), followed by Coimbatore (0.217) and Bhopal (0.210). Conversely, Kolkata (0.132), Mumbai (0.139), and Navi Mumbai (0.144) exhibited considerably lower average rooftop reflectivity.

These findings indicate that rooftop surface characteristics differ substantially among Indian cities and may reflect variations in construction materials, roofing practices, and urban morphology, although these factors were not explicitly evaluated in the present study.

Figure 3:

Distribution of Rooftop Albedo:

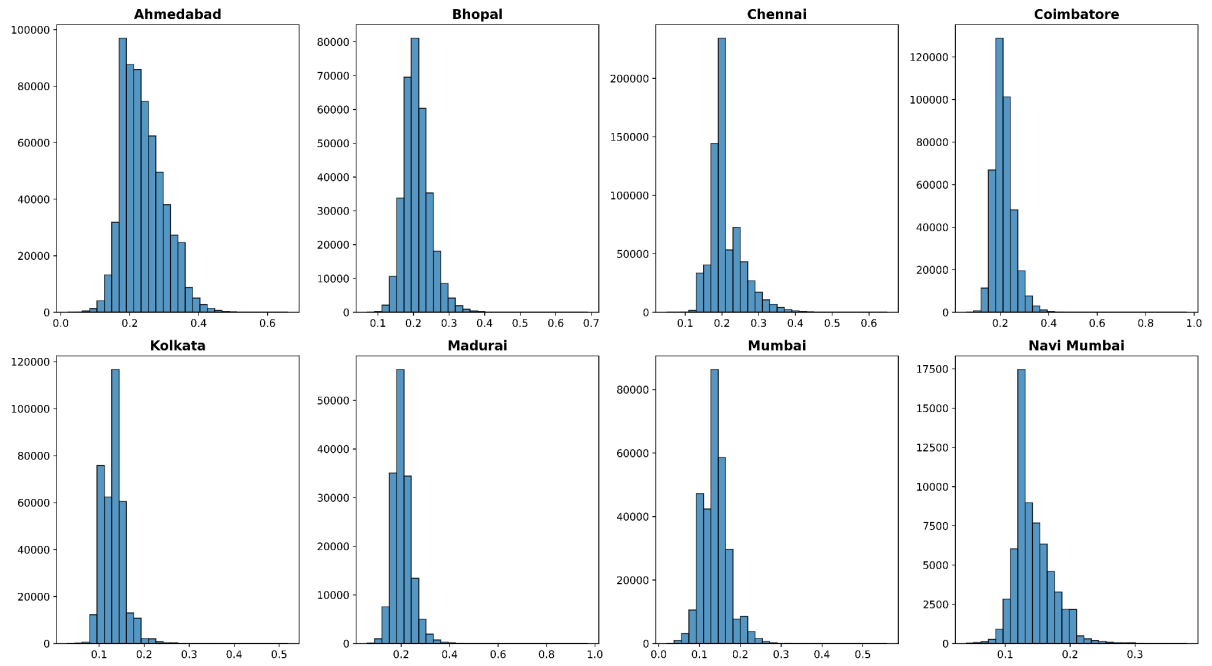


Interpretation:

The boxplots reveal differences in both the central tendency and variability of rooftop albedo across cities. Ahmedabad exhibits the highest median rooftop albedo, whereas Kolkata, Mumbai, and Navi Mumbai display consistently lower distributions. Several cities also contain high-value outliers, indicating the presence of highly reflective rooftop surfaces within otherwise darker urban environments.

Figure 4:

Histograms:



Interpretation:

The frequency distributions indicate that rooftop albedo values are generally unimodal but vary in spread and skewness among cities. Ahmedabad and Coimbatore exhibit broader distributions with a greater proportion of higher-albedo rooftops, whereas Kolkata and Mumbai are characterized by narrower distributions concentrated at lower reflectivity values. These differences further emphasize the diversity of rooftop characteristics among India's urban environments.

Table 1:

Ran k	City	Roofs	Mea n	Media n	SD	Low (%)	Moderate (%)	High (%)
1	Ahmedabad	616,972	0.239	0.23	0.060	24.0	61.7	14.4
2	Coimbatore	388,906	0.217	0.21	0.041	30.5	66.3	3.2
3	Bhopal	326,982	0.210	0.21	0.036	35.5	62.9	1.6

4	Chennai	694,444	0.209	0.20	0.044	44.2	51.9	3.8
5	Madurai	156,081	0.207	0.20	0.040	40.1	57.9	2.1
6	Navi Mumbai	64,329	0.144	0.14	0.031	94.5	5.5	0.1
7	Mumbai	301,525	0.139	0.14	0.032	95.0	4.9	0.1
8	Kolkata	357,569	0.132	0.13	0.025	98.5	1.5	0.0

A total of 2.91 million rooftop observations across eight Indian cities were retained for city-level analysis after spatial clipping. Ahmedabad exhibited the highest mean rooftop albedo (0.239), while Kolkata showed the lowest (0.132). More than 95% of rooftops in Mumbai and nearly 98.5% in Kolkata fall within the low-albedo category (<0.20), whereas Ahmedabad displayed the largest proportion of moderate-to-high albedo rooftops. H3 aggregation revealed pronounced neighborhood-scale variability in rooftop reflectivity across all cities, demonstrating that rooftop characteristics are spatially heterogeneous and cannot be adequately represented by city-wide averages alone.